



BITS Pilani

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EMPLOYMENT

Senior NLP Engineer - Contract role



Feb 24 – Present

FuturePath AI

- Built an Agent assist using LLMs (OpenAI GPT-4o, Llama 3.1, Anthropic sonnet) & Llamaindex with semantic routing to dynamically select appropriate agents (RAG, web search, ticketing, etc) for efficient resolution and reporting. The system automates responses, actions, improving user experience and reducing operational costs.
- Led the development of an AIOps platform that automated L1/L2 support tasks, help in L3 & assigned issues to appropriate resolver groups using AI-ML

Senior data scientist - GenAI



Jun 23 – Feb 24

Paytm

- Led the Copilot project to automate the generation of monthly reports and ad-hoc analyses.
- Developed an AI assistant utilizing multi-agent architecture and function calling to manage reporting & tasks with AWS-bedrock & Advanced RAG pipeline (Hybrid search, Re-ranking, RAFT, Tree & Graph based)
- Led the deployment, optimization, Fine-tuning, & monitoring of Large Language Models (LLMs) in production.

Data Scientist



Jun 21 – Jun 23

TOMTOM

- Working with MCP-Engineering to manage the real-time MAPs data flow.
- Implemented the auto-serving of task model using Large Language Models (LLM), Graph analytics & RAG for road edits. Which speeds up the process by more than 20x times while maintaining the same efficiency.
- Created a 360 view of MAP editors to predict asynchronous wait time between the tasks. This alert increases productivity by 28%
- Used Pyspark and Scala, to optimization of data pipelines on Azure Databricks
- Created a Customer Sentiment & Emotion Analysis model for product performance on various platforms & received the STAR AWARD (Best result)
- Working on LLMs (BERT, BART, GPT-3/4, Llama, and Mistral) with the help of LangChain to automatically resolve customers', clients', and stakeholders' queries and issues.
- With the Reporting team created dashboards for Business Stakeholders (Apple, Microsoft, Uber, etc.).

Data Scientist



Feb 20 – Jun 21

EC Infosolutions Pvt. Ltd.

- Assisted in scientific research on Animal Electrochemical Impedance Spectroscopy with advanced statistical analysis and BI Tools. Created predictive models to predict and identify the disease, pregnancy, ovulation, etc., in the animal at early stages with 82% accuracy.
- Conducted regression analysis (Time Series forecasting, Trend analysis and Statistical modeling) for patient footfall. Achieving a 21% more accurate prediction of performance than previous models.
- Create a pipeline on AWS Sagemaker to automate the process, resulting in a 39% increase in usability for non-technical staff members.
- Develop a custom shape predictor model from scratch to find and extract facial feature landmarks for dogs using Dlib, Opencv & Neural Networks (CNN)

FREELANCING

Data Scientist - Locobuzz



Jan 2022 – Nov 2022

- Created Sentiment, Forecast reach and Frequency models.
- With the help of prompt engineering and generative models, created a ticket categorization custom model & chatbot to find, categorize, and respond to the end customer based on their query, requests, etc., on various platforms
- Deploy all the models on AWS Sagemaker

Data Scientist - British American Tobacco



Oct 2021 – Jun 2022

- Worked with the analytics team to develop a model to predict tobacco yield delivered by the farmers based on weather and soil information
- Used Statical(A/B) testing & Time series analysis to identify the best-selling strategy & loss of potential sales due to a new competitor.
- To identify the best prices, gather data from various sources to define the price flexibility, demographic intelligence, competitor's price, seasonality, etc.
- Created end-to-end deployment pipeline on Azure and Snowflake

Sr. Data Science Manager - Motilal Oswal



Jun 2021 – Sep 2021

- Working with the Risk team to predict Cheque bounces and Credit behaviour over time using machine learning and Statical concepts
- Created a rule engine using deep learning & graph analytics to identify when one will become non-performing assets & the reason behind them (causality engine)

EDUCATION

M.Tech in Data Science And Engineering

Birla Institute of Technology & Science - BITS Pilani (Incomplete)

B.E in Information Technology

8.41/10

Vidyalankar Institute of Technology - University of Mumbai



2016 – 2020

PUBLICATIONS

Manpreet Sandhu, Pratik Hadawale, Saumaan Momin and Prof. Ajitkumar Khachane. "Plant Disease Detection using ML and UAV". IRJET journal. 2020 Mar;7(3):5352-57

SKILLS

- Programming Languages:** Python, JavaScript and R
- Tools:** SQL, AWS, Sagemaker, Git, Jupyter, Scikit-learn, Pandas, SciPy, Numpy, Psychopy, Matplotlib, Seaborn, Nltk, PySpark, GCP, OpenCV, StatsModels, Spacy, Hive, Redshift, ggplot2, Tableau, SHAP, LangChain, LlamaIndex, Vector DB.
- Frameworks:** PyTorch, TensorFlow, Keras, D3.js and Django
- Algorithms:** Regression, Classification, Clustering, Boosting(LGBM & XGboost), Bagging(Random forest), Naive Bayes, Back propagation, Gradient descent, Activation functions, Causality, LLM, Generative models & Transformers, Re-ranking.
- Techniques:** Machine learning, time series analysis, convolution neural networks, advanced statistical and predictive modeling, Object detection, Image segmentation, NLP, Recommender system and data visualization.

PROJECTS & CONTRIBUTION

Road Lane Detection Networks-for Autonomous Driving

- We transcend the aforementioned limitations and propose to cast the lane detection problem as an instance segmentation problem – during which each lane forms its instance – that can be trained end-to-end
- Our lane detection algorithm, running at 138 fps, which can handle a variable number of lanes and affect lane changes.

Python OpenCV Object detection Tensorflow segmentation

Medical specialty from Transcription

- The transcript tells us about the medical case, however, the medical term that refers to the complete transcript is the medical specialty & that is more close to the standard medical term.

Python NLP nltk Spacy Bag of words Word2Vec TF-IDF

Restaurants recommendation system

- Scrapping the data from different sources and then combining it to form a single-ton dataset
- Produces a list of top restaurants and the most common venue item that the user can enjoy

Python Clustering IBM Watson Scraping Recommender system

CERTIFICATIONS

- IBM Data Science Professional Certificate (Specialization)
- Oracle Autonomous Database Cloud 2019 Certified Specialist
- Google Analytics Individual Qualification
- Python from University of Michigan
- Machine Learning from Stanford University
- Maharashtra Talent Search Examination
- Data Engineering, Big Data & ML on GCP (Specialization)

INTERNSHIP

Data Engineer - Dell Technologies



May 2019 – Nov 2019

- Was part of the CSAT team and aided in developing an application that automates the ETL process.
- Performed Data Migration from Amazon S3 to Amazon Redshift.
- Worked on integrating the Hive data warehouse system with other data sources such as Amazon Redshift.